

"Library Functions Manual"

NAME

saw – function

LIBRARY

ciaaw library (*libciaaw*, *-lciaaw*)

SYNOPSIS

```
Fortran      real(dp) function saw(s,ab,u)result(res)  
              character(len=*), intent(in) :: s  
                Element symbol.  
              logical, intent(in), optional :: ab  
                Set to False if the abridged value is not desired. Default to TRUE.  
              logical, intent(in), optional :: u  
                Set to True if the uncertainty is desired. Default to FALSE.  
              real(dp) :: res  
                NaN if the provided element is incorrect or -1 if the element does not have a SAW.  
  
C            double ciaaw_saw(char *s,int n,bool ab,bool u);  
  
Python      saw(s:str,abridged:bool=True,uncertainty:bool=False)->float:
```

DESCRIPTION

Get the standard atomic weight for the element s.

RETURN VALUE

NaN If the provided element is incorrect.
-1 If the element does not have a SAW.

EXAMPLE

```
Fortran:  
      use ciaaw  
      saw("H", ab=.false., u=.true.)  
  
C:  
      include <stdio.h>  
      include "ciaaw.h"  
      ciaaw_saw("H", 1, false, true)  
  
Python:  
      import pyciaaw  
      pyciaaw.saw("H", ab=False, u=True)
```

SEE ALSO

***ciaaw*(3), *ciaaw_cli*(1), *ciaaw_version*(3), *ciaaw_types*(3)**